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NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (CSE) (Credit System) Degree Examinations
Supplementary Examinations - December 2025

21CS702 / 20CSE54 - CRYPTOGRAPHY AND NETWORK SECURITY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Cryptography. Demonstrate Playfair cipher algorithm and using Playfair cipher algorithm encrypt the plaintext "BALLOON" with Key "MONARCHY".	10	L*2	1	1,2
	b) Demonstrate the HILL cipher with an example.	10	L2	1	1
2.	a) Demonstrate the Feistel decryption algorithm with an example.	10	L2	2	1
	b) With neat diagram describe the general depiction of DES encryption Algorithm.	10	L2	2	1,2
3.	a) Explain Substitution and transposition techniques with an example.	6	L2	1	1
	b) Explain the block cipher modes of operation.	4	L2	2	1,2
	c) Describe the blowfish encryption and decryption works with a neat diagram.	10	L2	2	1,2
Unit – II					
4.	a) Explain the public-key cryptography with a neat diagram.	10	L2	3	1,2
	b) Show the generation of keys in RSA algorithm. Perform encryption and decryption using RSA algorithm for the following P=7; q=11; e=13 and M=8.	10	L3	3	1,2
5.	a) Differentiate between conventional encryption and public-key encryption. Also explain the public-key cryptosystem in regards to secrecy and authentication.	10	L2	3	1,2
	b) With a neat diagram, explain the digital signature algorithm.	10	L2	3	1
6.	a) Explain the benefits of IPsec and IP security scenario with a diagram.	10	L2	4	1,2
	b) Describe the Diffie-Hellman key exchange algorithm with an example.	10	L2	3	1,2
Unit – III					
7.	a) Define SET. Describe the business requirements for SET, its key features, and the participants in SET transactions.	10	L2	5	1,2
	b) Explain SSL architecture and discuss the various operation of the SSL record protocol.	10	L2	5	1
8.	a) Describe the taxonomy of malicious programs. Explain the four phases of typical virus.	10	L2	5	1,2
	b) Define firewall. Discuss the firewall types with a neat diagram.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
